

UFCEP6-0-3 - Project & Test Plan

Hospital Patient Referral, Treatment and Administration System Initial Release - 28 September 2026

1. Project Scope

The Initial Release implements an integrated hospital patient pathway using executable BPMN, Camunda Forms and external workers. The workflow covers referral and consultant review, appointment and patient communication, treatment/chemotherapy, funding/payment/refund, and enquiry/clinic-letter/pathway monitoring. External scheduling, correspondence, treatment-resource and payment services are simulated for the coursework demonstration.

2. Team Ownership

First Owner	Primary Area	Second Owner
DENG JIN WANG	Referral & Consultant Review	WU JIA WEI
WU JIA WEI	Appointment & Patient Communication	LIN JIA QIANG
LIN JIA QIANG	Treatment & Chemotherapy	WU TONG YU
WU TONG YU	Funding / Payment / Refund	XIAO YU YING
XIAO YU YING	Enquiry / Clinic Letter / Pathway Monitoring	DENG JIN WANG

3. Sprint Goals

Sprint	Goal
Sprint 1	Establish requirements, ownership, initial executable BPMN increment, repository structure and configuration-management evidence.
Sprint 2	Complete forms, external workers, gateway variables/conditions, integration, testing and a demonstrable Initial Release.

4. Prioritised Product Backlog

ID	Priority	Backlog Item	Owner	Dependency
PB-01	Must	Receive/record referral and supporting information	DENG	Referral source available
PB-02	Must	Check completeness and request missing information	DENG	PB-01
PB-03	Must	Consultant Accept / Reject / More Information / Redirect with reason	DENG	PB-02
PB-04	Must	Prevent appointment before authorised acceptance	DENG	PB-03

ID	Priority	Backlog Item	Owner	Dependency
PB-05	Must	Search external scheduling service for suitable slot	WU JIA WEI	PB-03
PB-06	Must	Confirm appointment and send appointment letter	WU JIA WEI	PB-05
PB-07	Must	If appointment within two weeks, also require telephone contact	WU JIA WEI	PB-06
PB-08	Must	Record contact outcomes: answered/no answer/wrong number/reschedule	WU JIA WEI	PB-07
PB-09	Must	Handle no-slot and urgent-referral escalation path	WU JIA WEI	PB-05
PB-10	Must	Record consent and authorised treatment booking request	LIN	Appointment/consultation
PB-11	Must	Reject incomplete/unauthorised treatment request	LIN	PB-10
PB-12	Must	Check treatment/lab/imaging availability and avoid duplicate bookings	LIN	PB-10
PB-13	Must	Clinical Continue / Delay / Modify decisions remain authorised clinical tasks	LIN	PB-12
PB-14	Must	Determine hospital/insurer/patient funding route	WU TONG YU	Treatment request
PB-15	Must	Secure simulated payment request and status handling	WU TONG YU	PB-14
PB-16	Must	Investigate confirmation loss/duplicate risk instead of recharging	WU TONG YU	PB-15
PB-17	Must	Process authorised refund and record result	WU TONG YU	Financial impact
PB-18	Must	Record/classify/prioritise/route enquiries	XIAO	Patient enquiry
PB-19	Must	Prevent call handlers from providing clinical advice	XIAO	PB-18

ID	Priority	Backlog Item	Owner	Dependency
PB-20	Must	Create/approve/ admin-check/ distribute Clinic Letter	XIAO	Consultation complete
PB-21	Must	7-day monitoring, weekly reminders, >1 month and >3 month escalation	XIAO	PB-20
PB-22	Must	Authentication/ RBAC/audit/data protection requirements represented in design evidence	Group	All areas
PB-23	Should	Accessible communication/trans lation/representative support documented	Group	Communication tasks
PB-24	Must	Executable BPMN, Forms, Workers, deployment configuration and test evidence	Group	All

5. Sprint Backlog and Status

ID	Activity	First	Second	Estimate	Dependency	Acceptance Condition	Status
S1-01	Referral requirements + BPMN	DENG	WU JIA WEI	6h	Case study	Referral routes represented	Done
S1-02	Appointment requirements + BPMN	WU JIA WEI	LIN	7h	Accepted referral	Booking/ contact routes represented	Done
S1-03	Treatment requirements + BPMN	LIN	WU TONG YU	7h	Confirmed appointment	Consent/ authorisation/resource routes represented	Done
S1-04	Funding/ payment/ refund BPMN	WU TONG YU	XIAO	7h	Treatment request	Funding/ payment/ refund routes represented	Done
S1-05	Enquiry/ clinic-letter BPMN	XIAO	DENG	7h	Consultation info	Enquiry/ monitoring/ escalation represented	Done
S1-06	Integrated executable BPMN + gateway conditions	Group	Group	6h	S1-01..05	XML valid; executable flag; worker bindings; conditions	Done

ID	Activity	First	Second	Estimate	Dependency	Acceptance Condition	Status
S2-01	Referral & Consultant Forms	DENG	WU JIA WEI	4h	S1-01	Forms bound to user tasks	Done
S2-02	Appointment Forms + scheduling/correspondence workers	WU JIA WEI	LIN	6h	S1-02	Forms + two appointment workers present	Done
S2-03	Treatment Forms + resource worker	LIN	WU TONG YU	5h	S1-03	Treatment form + worker present	Done
S2-04	Funding Form + payment/refund workers	WU TONG YU	XIAO	6h	S1-04	Funding form + two workers present	Done
S2-05	Enquiry/ Clinic Letter Forms + distribution worker	XIAO	DENG	6h	S1-05	Forms + correspondence worker present	Done
S2-06	Static configuration validation	Group	Group	2h	S2-01..05	BPMN/ form/ worker bindings validated	Done
S2-07	Worker component tests	Group	Group	2h	Worker logic	7 local component tests pass	Done
S2-08	Camunda runtime integration test and demo rehearsal	Group	Group	4h	Teaching Camunda environment	Normal + exception scenario executed with screenshots/ logs	To execute in final environment

6. Definition of Done

- Artefact/component exists in the shared repository.
- Agreed acceptance condition is met.
- First owner completed the work and second owner can explain/continue it.
- The component has been reviewed and is linked/integrated where applicable.
- Configuration/version is identifiable.
- Known limitations and assumptions are recorded.
- Relevant tests have been executed and evidence retained.

7. Configuration Management

All BPMN, Forms, worker source, plans, tests and evidence are kept in the shared Git repository. The submitted version must be identified by an actual Git tag or commit hash in SUBMITTED_VERSION.txt. Small commits should be used for traceability, and a release snapshot should be preserved for SR1/SR2 and the final Initial Release.

8. Risks and Responses

ID	Risk	Probability	Impact	Response
R-01	BPMN cannot deploy/execute	Medium	High	Validate XML/configuration early; execute final deployment rehearsal
R-02	Gateway variable mismatch	Medium	High	Use shared variable names and static validation
R-03	Form binding/key mismatch	Medium	High	Validate every BPMN formId against editable .form files
R-04	Worker/service failure	Medium	High	Retries + failure messages; simulate failure/alternative paths
R-05	Duplicate payment/booking	Low	High	Investigation route; avoid automatic repeat charge/duplicate appointment
R-06	Clinical/admin responsibility crossed	Low	High	Keep clinical decisions with authorised clinical roles
R-07	Integration delayed	Medium	High	Continuous integration and second-owner handover
R-08	Case-study ambiguity	Medium	Medium	Record assumption and obtain tutor clarification

9. Test Strategy

Testing is performed at four levels: (1) static configuration validation of BPMN/Form/Worker bindings, (2) worker component testing of simulated external-service logic, (3) process integration testing in Camunda, and (4) end-to-end demonstration scenarios. Normal, alternative, exception and relevant non-functional behaviour are covered.

10. Detailed Test Plan

ID	Area	Scenario	Test Data / Preconditions	Expected Outcome
T01	Referral	Complete referral	informationComplete=true	Consultant review becomes available

ID	Area	Scenario	Test Data / Preconditions	Expected Outcome
T02	Referral	Missing information	informationComplete=false	Missing-information request path; returns for re-check
T03	Referral	Consultant accepts	referralDecision=ACCEPT	Appointment booking begins
T04	Referral	Reject/redirect/more information	Each decision value	Correct alternative path selected
T05	Appointment	Suitable slot	requestedSlotAvailable=true	slotAvailable=true; appointment may be confirmed
T06	Appointment	No suitable slot	requestedSlotAvailable=false	No-slot/pathway route selected
T07	Appointment	Appointment <=14 days	Date +7 days	appointmentWithinTwoWeeks=true; telephone required
T08	Appointment	Contact outcomes	ANSWERED/ NO_ANSWER/ WRONG_NUMBER/ RESCHEDULE	Correct path and result recorded
T09	Treatment	Patient consents	patientConsent=true	Treatment request proceeds
T10	Treatment	Unauthorised request	treatmentAuthorised=false	Request returned/rejected
T11	Treatment	Resource unavailable	requestedTreatmentAvailable=false	Pending + notify/retry path
T12	Treatment	Continue/Delay/Modify	clinicalDecision values	Correct clinical route
T13	Funding	Hospital/insurer/patient routes	fundingType values	Correct funding branch
T14	Payment	Successful payment	paymentScenario=SUCCESSFUL	SUCCESSFUL + transaction reference
T15	Payment	Declined/cancelled/incomplete	Scenario values	Retry/notification route
T16	Payment	Confirmation lost	CONFIRMATION_LOST	Investigation required; no automatic recharge
T17	Refund	Refund required	refundRequired=true	Approved refund request result recorded
T18	Enquiry	Admin/finance/clinical	enquiryType values	Correct team route
T19	Enquiry	Urgent clinical concern	urgentClinicalConcern=true	Immediate clinical escalation
T20	Clinic Letter	Not clinically approved	clinicLetterApproved=false	Distribution worker rejects input
T21	Clinic Letter	Completed within 7 days	clinicLetterWithin7Days=true	No delayed-letter monitoring
T22	Clinic Letter	Delayed	clinicLetterWithin7Days=false	Monitoring/reminder path
T23	Clinic Letter	>1 month / >3 months	letterAgeBand	Correct escalation level
T24	Configuration	BPMN/form/worker linkage	Release files	All service job types, forms and conditions

ID	Area	Scenario	Test Data / Preconditions	Expected Outcome
				resolve
T25	End-to-End	Normal patient journey	Accepted referral + slot + consent + funding + letter	Process reaches completion
T26	End-to-End	Exception journey	No slot/payment issue/delayed letter	Relevant exception route demonstrated

11. Testing Conducted - Evidence

The following tests were actually executed locally against this release package. These results are not fabricated; the raw output is retained in 05_Test_Evidence.

Evidence ID	Test	Result	Actual Result	Evidence File
EV-01	Static release validation	PASS	BPMN parses; isExecutable=true; 6 service-worker job types; 10 form bindings; 51 condition expressions; 9 valid form files.	05_Test_Evidence/static_configuration_validation.txt
EV-02	Worker logic component test	PASS	7/7 tests passed: appointment two-week rule, no slot, payment success, confirmation-lost investigation, unavailable resource, Clinic Letter approval, refund.	05_Test_Evidence/worker_component_test_output.txt

12. Runtime Tests Still Required Before Final Blackboard Submission

The local environment used to prepare this package does not contain the group's authorised Camunda teaching cluster. Therefore the following must be executed in the group's actual Camunda environment before the final demonstration: deploy BPMN/forms, run at least one complete normal instance, run at least one alternative/failure instance, capture Tasklist/Operate screenshots or logs, and record the actual Git commit/tag used. Do not replace these with invented results.

13. Known Assumptions / Tutor Clarification

- Exact urgent-referral criteria are not defined in the case study.
- Exact telephone retry count before escalation is not specified.
- Exact external-service timeout/retry policy is not specified.
- Exact urgent-clinical-enquiry rules are incomplete.
- External scheduling, correspondence, treatment-resource and payment services are simulated in this Initial Release.

14. Initial Release Deliverable Map

Assessment Deliverable	Package Location
Operational BPMN + deployment configuration	01_Operational_Model + 07_Configuration
External workers + dependencies/config templates	02_External_Workers + 07_Configuration
Editable Camunda Forms + task bindings	03_Camunda_Forms + BPMN form bindings
Project & Test Plan + test evidence	04_Project_Test_Plan + 05_Test_Evidence
Working Initial Release demonstration support	06_Demonstration + submitted repository version